Stawberries 3

MA615

2024 Sept 30

```
{r setup, include=FALSE} knitr::opts_chunk$set(echo = TRUE)
```

Version 6

We ditch the counties

Preparing data for analysis

Acquire, explore, clean & structure, EDA

Data cleaning and organization

“An introduction to data cleaning with R” by Edwin de Jonge and Mark van der Loo

“Problems, Methods, and Challenges in Comprehensive Data Cleansing” by Heiko Müller and Johann-Christoph Freytag

Strawberries

Questions

Where they are grown? By whom?

Are they really loaded with carcinogenic poisons?

Are they really good for your health? Bad for your health?

Are organic strawberries carriers of deadly diseases?

When I go to the market should I buy conventional or organic strawberries?

Do Strawberry farmers make money?

How do the strawberries I buy get to my market?

The data

The data set for this assignment has been selected from:

[USDA_NASS_strawb_2024SEP25 The data have been stored on NASS here: USDA_NASS_strawb_2024SEP25

and has been stored on the blackboard as strawberries25_v3.csv.

read and explore the data

Set-up

```{r} #| label: load libraries and set options #| warning: false #| message: false #| echo: false

library(knitr)
library(kableExtra) library(tidyverse)

```
Read the data and take a first look

```{r}
#| label: read data - glimpse

strawberry <- read_csv("strawberries25_v3.csv", col_names = TRUE, show_
col_types = FALSE )

## glimpse(strawberry)
```

Ditch the counties

```{r} #| label: ditch the counties

unique(strawberry$Geo Level)

strawberry <- strawberry |> filter(Geo Level== "NATIONAL" | Geo Level== "STATE")

```
I have 5359 rows and 21 columns.

remove columns with a single value in all rows

```{r}
#|label: function def - drop 1-item columns

drop_one_value_col <- function(df){
 # browser()
  df_id <- ensym(df)
  msg = paste("Looking for single value columns in data frame: ",as.cha
racter(df_id) )
  print(msg)
  ## takes whole dataframe
dropc <- NULL
val <- NULL
## test each column for a single value
for(i in 1:dim(df)[2]){
  if(dim(distinct(df[,i]))[1] == 1){
```

```
    dropc <- c(dropc, i)
    val <- c(val, df[1,i])
  }
}

if(is.null(dropc)){
  print("No columns dropped")
  return(df)}else{
   print("Columns dropped:")
   # print(colnames(df)[drop])
   print(unlist(val))
   df <- df[, -1*dropc]
   return(df)
   }
}


## use the function

strawberry <- strawberry |> drop_one_value_col()
```

Split the census data from the survey data. drop single value columns

Census data first

```{r} #| label: srawberries split census, survey

straw_cen <- strawberry |> filter(Program==“CENSUS”)

straw_sur <- strawberry |> filter(Program==“SURVEY”)

straw_cen <- straw_cen |> drop_one_value_col()

straw_sur <- straw_sur |> drop_one_value_col()

```{r}
#| label: straw_cen split cols

straw_cen <- straw_cen |>
  separate_wider_delim(  cols = `Data Item`,
                         delim = " - ",
                         names = c("strawberries",
                                   "Category"),
                         too_many = "error",
                         too_few = "align_start"
                       )
```

```{r} #| label: isolate organic

## unique(straw_cen$strawberries)  # straw_cen$strawberries |> str_which("STRAWBERRIES") |> length()

## straw_cen$strawberries |> str_which("STRAWBERRIES, ORGANIC") |> length()  # straw_cen$strawberries |> str_which("STRAWBERRIES, ORGANIC, FRESH MARKET") |> length()

## straw_cen$strawberries |> str_which("STRAWBERRIES, ORGANIC, PROCESSING" ) |> length()  #  # ## count the cases  # straw_cen$strawberries |> str_which("ORGANIC") |> length()

## straw_cen$strawberries |> str_which("FRESH MARKET") |> length()  # straw_cen$strawberries |> str_which("PROCESSING") |> length()

straw_cen <- straw_cen |> separate_wider_delim( cols = strawberries, delim = ",", names = c("strawberries", "ORGANIC", "organic_detail"),

```
                      too_many = "error",
                      too_few = "align_start"
)
```

straw_cen <- straw_cen |> drop_one_value_col()

how many organic rows?

organic_cen <- straw_cen |> filter(ORGANIC == "ORGANIC")

sum(is.na(straw_cen$ORGANIC))

straw_cen <- straw_cen[(is.na(straw_cen$ORGANIC)),]

straw_cen <- straw_cen |> drop_one_value_col()

```
Note that straw_cen has only one year: 2022
```

```
Current stats Census date has been isolated and split between Organic a
nd Conventional strawberries

```{r}
#| label: explore straw_cen$Domain Category


straw_cen <- straw_cen |>
  separate_wider_delim(  cols = `Domain Category`,
                         delim = " ",
                         names = c("COL1",
                                   "COL2"),
                         too_many = "merge",
                         too_few = "align_start"
                       )


straw_cen$COL2 <- str_replace(straw_cen$COL2,"WITH ","")

straw_cen <- straw_cen |> rename(Measure = COL1, Bearing_type= COL2)
```

```{r} #| label: explore straw_cen$Domain & Domain Category

## remove AREA GROWN and parens

## change NOT SPECIFIEC TO TOTAL

straw_cen <- straw_cen |> rename(size_bracket = `Domain Category`)

straw_cen$size_bracket <- str_replace(straw_cen$size_bracket, "NOT SPECIFIED", "TOTAL")

straw_cen$size_bracket <- str_replace(straw_cen$size_bracket, "AREA GROWN:", "")

```{r}
#| label: explore organic_cen

organic_cen <- organic_cen |> drop_one_value_col()
```

Now examine the Survey data

```{r} #| label: Survey data
```

Data Item

unique(straw_sur$Data Item)

straw_sur1 <- straw_sur |> separate_wider_delim(cols = Data Item, delim = ",", names = c("straw", "mkt", "measure", "other"), too_many = "merge", too_few = "align_start")

straw_sur2 <- straw_sur1 |> separate_wider_delim(cols = "straw", delim = " - ", names = c("straw", "more"), too_many = "merge", too_few = "align_start")

rm(straw_sur, straw_sur1)

```
Shift data into alignment

```{r}
#| label: Shift data on a row

## function shift_loc
## Moves adjacent data cells in a data.frame on a single row
## Use this function to fix alignment problems after separating
## columns containing multiple columns of data.

## Of course the working assumption is that there is room in the
## data frame for the data you're shifting.
##
## The data cells that are empty after the data shift are NA.
##
## Input paramaters
##
## df -- data frame
## col_name -- name of colume where the left-most data item is located
## dat_name -- name of data item in the column
## num_col -- the number of columns is the same as the number of
##            adjacent data to be moved.
## num_shift -- the number of rows to move the data
##


shift_loc <- function(df, col_name, dat_name, num_col, num_shift){
 # browser()
  col_num = which(colnames(df) == col_name)
  row_num = which(df[,col_num] == dat_name)  ## calcs a vector of rows

  for(k in 1:length(row_num)){
  d = rep(0,num_col) ## storage for items to be moved
  for(i in 1:num_col){
    d[i] = df[row_num[k], col_num + i - 1]
```
```

```
  }
  for(i in 1:num_col){
    ra = row_num[k]
    cb = col_num + i - 1
    df[ra, cb] <-  NA
  }
  for(j in 1:num_col){
    rc = row_num[k]
    cd = col_num + j - 1 + num_shift
    df[rc, cd] = d[j]
  }
  }
 # sprintf("Rows adjusted:")
  # print("%d",row_num)
  return(df)
}




straw_sur2 <- straw_sur2 |> shift_loc("more", "PRICE RECEIVED", 2, 1 )


straw_sur2 <- straw_sur2 |> shift_loc("more", "ACRES HARVESTED", 1, 1 )

straw_sur2 <- straw_sur2 |> shift_loc("more", "ACRES PLANTED", 1, 1 )

straw_sur2 <- straw_sur2 |> shift_loc("more", "PRODUCTION", 2, 1 )

straw_sur2 <- straw_sur2 |> shift_loc("more", "YIELD", 2, 1 )

straw_sur2 <- straw_sur2 |> shift_loc("more", "APPLICATIONS", 3, 1 )

straw_sur2 <- straw_sur2 |> shift_loc("more", "TREATED", 3, 1 )

straw_sur2 <- straw_sur2 |> drop_one_value_col()
```

Examine Domain

```{r} #| label: use domain to make chem, fert, total data frames

unique(straw_sur2$Domain)

straw_sur2 <- straw_sur2 |> separate_wider_delim(cols = Domain, delim = ",", names = c("col1", "col2"), too_many = "merge", too_few = "align_start")

unique(straw_sur2$col1)

survey_d_total <- straw_sur2 |> filter(col1 == "TOTAL")
```

survey_d_chem <- straw_sur2 |> filter(col1 == “CHEMICAL”)

survey_d_fert <- straw_sur2 |> filter(col1 == “FERTILIZER”)

```
now look at totals

```{r}


survey_d_total <- survey_d_total |> drop_one_value_col()

### align terms



survey_d_total <- survey_d_total |> shift_loc("measure", "MEASURED IN
$ / CWT", 1, 1 )


survey_d_total <- survey_d_total |> shift_loc("measure", "MEASURED IN
$", 1, 1 )


survey_d_total <- survey_d_total |> shift_loc("measure", "MEASURED IN C
WT", 1, 1 )

survey_d_total <- survey_d_total |> shift_loc("measure", "MEASURED IN T
ONS", 1, 1 )


survey_d_total <- survey_d_total |> shift_loc("measure", "MEASURED IN C
WT / ACRE", 1, 1 )

survey_d_total <- survey_d_total |> shift_loc("measure", "MEASURED IN T
ONS / ACRE", 1, 1 )


#### split the mkt column


survey_d_total <- survey_d_total |>
  separate_wider_delim(cols = mkt,
                       delim = " - ",
                       names = c("col3",
                                 "col4"),
```
```

```
                    too_many = "merge",
                     too_few = "align_start")
```

there are two markets for Strawberries – Fresh Marketing and Processing

make a table for each

from the Survey Totals

we have reports for

Markets: Fresh and Processing Operations: Growing and Production

```{r}

survey_d_total <- survey_d_total |> group_by(Year) |> group_by(Period) |> group_by(Geo Level) |> group_by(State) |> group_by(col3)

```{r}

survey_d_total_ca <- survey_d_total |>
  filter(State == "CALIFORNIA")

ca_tab <- survey_d_total_ca |> group_by(Year, Period
                                        )

##

ca_tab_22 <- survey_d_total_ca |> filter(Year == 2022)

ca_tab_22 <- ca_tab_22 |> drop_one_value_col()

ca_tab_22 <- ca_tab_22 |>
  filter(Period == "YEAR")

ca_tab_22 <- ca_tab_22 |>
  filter(Value != "(D)")

ca_tab_22 <- ca_tab_22 |> drop_one_value_col()
```

```{r} #| label: chemicals 2022

ca_chem_22 <- survey_d_chem |>
filter(Year == 2021) |> filter(State == “CALIFORNIA”)

ca_chem_22 <- ca_chem_22 |> drop_one_value_col()

The codes I use here form a summary of the top 10 states with the highe
st strawberries production. This answer the first question, which asks
where are the strawberries growing.

```{r}
# Q1

strawberry <- read_csv("strawberries25_v3.csv", col_names = TRUE, show_
col_types = FALSE )

# Create a summary of strawberry production by State and County
state_county_summary <- strawberry %>%
  group_by(State, County) %>%
  summarise(Count = n()) %>%
  arrange(desc(Count))

# Display the summary table
print(state_county_summary)

state_summary <- state_county_summary %>%
  group_by(State) %>%
  summarise(Count = sum(Count)) %>%
  arrange(desc(Count))

# Plot the top 10 states by strawberry production count
top_states <- head(state_summary, 10)

ggplot(top_states, aes(x = reorder(State, Count), y = Count)) +
  geom_bar(stat = "identity", fill = "orange") +
  coord_flip() +
  labs(title = "Top 10 States by Strawberry Production",
       x = "State", y = "Production Count") +
  theme_minimal()
```

The code here provides a list of the chemical data mainly used on strawberries. We can change the number in top_chemicals variable to see the name of the chemical and then check whether or not there are carcinogenic poisons.

```{r} # Q2
```

Filter rows that contain information about chemicals used

chemical_data <- strawberry %>% filter(grepl("CHEMICAL", `Domain Category`)) %>% select(State, County, `Domain Category`)

Display the chemical data

print(head(chemical_data))

Summarize the use of different chemicals

chemical_summary <- chemical_data %>% group_by(`Domain Category`) %>% summarise(Count = n()) %>% arrange(desc(Count))

Plot the most common chemicals used on strawberries

top_chemicals <- head(chemical_summary, 10)

ggplot(top_chemicals, aes(x = reorder(`Domain Category`, Count), y = Count)) + geom_bar(stat = "identity", fill = "red") + coord_flip() + labs(title = "Top 10 Chemicals Used on Strawberries", x = "Chemical Type", y = "Count of Usage") + theme_minimal()

```
Question 3 asks whether the strawberries are good for our health or not.
 I think the Domain.Category column in strawberries25 can determine whe
ther or not it is health, because organic fruits do not use pesticides
during the cultivation process. Base on the plot, most of strawberries
are organic, which are health.

```{r}
# Q3
# Filter organic and non-organic data
organic_data <- strawberry %>%
  filter(grepl("ORGANIC STATUS: \\(NOP USDA CERTIFIED\\)", `Domain Cate
gory`))

# Count the number of organic vs. non-organic data entries
organic_count <- nrow(organic_data)
non_organic_count <- nrow(strawberry) - organic_count

# Create a data frame for visualization
organic_summary <- data.frame(
  Type = c("Organic", "Non-Organic"),
  Count = c(organic_count, non_organic_count)
)
```
```

```
# Plot organic vs. non-organic production
ggplot(organic_summary, aes(x = Type, y = Count, fill = Type)) +
  geom_bar(stat = "identity") +
  labs(title = "Comparison of Organic vs. Non-Organic Strawberry Produc
tion",
       x = "Production Type", y = "Count") +
  theme_minimal()
```

There may be some columns relate to both the organic and deadly diseases to answer question 4, but I can't figure it out. As for question 5, I think we should buy organic strawberries since all the status of the organic strawberries are NOP USDA CERTIFIED.

```{r} # Q4 library(dplyr)

## Filter the data to only include rows related to organic strawberries

organic_strawberries <- strawberry %>% filter(grepl("ORGANIC STATUS: \(NOP USDA CERTIFIED\)", `Domain Category`))

## Check for diseases or harmful chemicals

diseases_info <- organic_strawberries %>% filter(grepl("DISEASE|PATHOGEN", tolower(`Domain Category`)))

## View the results

print(diseases_info)

## Q5

## Filter the data to only include rows related to organic strawberries

organic_strawberries <- strawberry %>% filter(grepl("ORGANIC STATUS: \(NOP USDA CERTIFIED\)", `Domain Category`))
```

Print the resulting data

print(organic_strawberries)

Alternatively, you can view the data in a more interactive way using:

View(organic_strawberries)

```
Question 6 asks do strawberry farmers make money. We choose the Data It
em column to observe. Based on the number of operations listed in the p
lot, we can say that strawberry farmers make money in some states of US.

```{r}
# Q6
# Filter data related to operations with strawberries
operation_data <- strawberry %>%
  filter(grepl("OPERATIONS", `Data Item`))

# Summarize the number of operations
operation_summary <- operation_data %>%
  group_by(State) %>%
  summarise(Operations = n()) %>%
  arrange(desc(Operations))

# Plot the number of operations by state
top_operations <- head(operation_summary, 10)

ggplot(top_operations, aes(x = reorder(State, Operations), y = Operatio
ns)) +
  geom_bar(stat = "identity", fill = "blue") +
  coord_flip() +
  labs(title = "Number of Strawberry Farming Operations by State",
       x = "State", y = "Number of Operations") +
  theme_minimal()
```

Now what I did is splitting the chemical data into use, name and code. The new data has three new columns with the chemical data so that these data are displayed clearly.

```{r} # Install and load dplyr if (!require("dplyr")) install.packages("dplyr") library(dplyr)

## Add new columns, and the new columns are all the chemical data in the Domain Category column.

strawberries <- strawberry %>% mutate( use = ifelse(grepl("^CHEMICAL,", Domain Category), sub("^CHEMICAL, (.*?):.", "\1", Domain Category), NA), name = ifelse(grepl(": \((.*) = ", Domain Category), sub("^CHEMICAL, .: \((.*?) = .\)$", "\\1", `Domain Category`), NA), code = ifelse(grepl("= (\\d+)\\)$", Domain Category), sub(".*= (\d+)\)$","\1", Domain Category), NA) )

## Display the updated data frame

head(strawberries)

```
```{r}
# Load necessary libraries
library(tidyverse)

# Load the data (replace 'strawberries.csv' with your file path)
library(readr)
strawberries <- read_csv("C:/Users/Dgy49137/Desktop/strawberries.csv")
View(strawberries)

# Convert "Value" and "CV (%)" to numeric, forcing non-numeric values t
o NA
strawberries$Value_numeric <- as.numeric(strawberries$Value)
strawberries$CV_numeric <- as.numeric(strawberries$`CV (%)`)

# Prepare the data: remove rows where both Value and CV are NA
df_clean <- strawberries %>%
  filter(!is.na(Value_numeric) | !is.na(CV_numeric))

# Predict missing values for 'Value_numeric' and 'CV_numeric'
missing_values <- strawberries %>%
  filter(is.na(Value_numeric) | is.na(CV_numeric))

# Check how many rows need prediction
num_missing_value <- sum(is.na(strawberries$Value_numeric))
num_missing_cv <- sum(is.na(strawberries$CV_numeric))

# Predict Value_numeric for missing rows
predicted_value <- predict(model_value, newdata = missing_values)

# Ensure the length of the predictions matches the number of missing ro
ws
if (length(predicted_value) == num_missing_value) {
```
```

```
  strawberries$Value_numeric[is.na(strawberries$Value_numeric)] <- pred
icted_value
} else {
  warning("The number of predicted 'Value_numeric' does not match the m
issing entries.")
}

# Predict CV_numeric for missing rows
predicted_cv <- predict(model_cv, newdata = missing_values)

# Ensure the length of the predictions matches the number of missing ro
ws
if (length(predicted_cv) == num_missing_cv) {
  strawberries$CV_numeric[is.na(strawberries$CV_numeric)] <- predicted_
cv
} else {
  warning("The number of predicted 'CV_numeric' does not match the miss
ing entries.")
}

# View the updated dataframe
print(strawberries)
```